

Evaluating The Pollution Patterns In Clear Lake

https://github.com/leozhangzzgx/Zhang_ENV872_Project_sp26.git

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Rationale and Research Question

Located within Lake County, Clear Lake is a popular destination in Northern California. Visitors, who may obtain a temporary relief for the hustle of Bay Area, enjoy boating, fishing, or hiking. With a history of two million years, Clear Lake is the one of the oldest lakes in North America (Molanphy). Volcanism has been playing a crucial role in the evolution of Clear Lake. It has produced volcanic ash that is very beneficial for the growth of grapes. It has also contributed to The Geysers, the largest geothermal power field on Earth (Pratt).

However, Clear Lake faces an uncertain future. Its freshwater ecosystem has been severely threatened by eutrophication. In 2017, a dedicated committee was endorsed by state legislation to boost the protection. Experts from UC Davis currently proposed the installation of a hypolimnetic oxygenation system (Hollis). In addition, a CDC report indicated that, although nearby private wells were generally safe, 58 percent of the private water systems directly utilizing lake water have been subject to excessive microcystin, a toxin generated by cyanobacteria. The increasing frequency of harmful algal blooms in Clear Lake might imply a formidable challenge to public health (Solomon et al.).

We decided to collect data from environmental monitoring. Focusing on a time series, we explored the tendency of different indicators of pollution. Under the pursuit for deep comprehension, we also explored the potential association among different indicators of pollution.

Dataset Information

From 2020 to 2025, scientists from California Water Science Center, a subsidiary of USGS, conducted a field study in Clear Lake. All data, in CSV format, could be downloaded from the official USGS website. The data should be employed under CC0-1.0 regulations (Bennett et al.).

Data Wrangling

The researchers used ‘-’ to represent missing values. In that case, we needed to manually modify the class of each column from string to numeric. For columns with only two unique values, we adopted the Boolean format. To avoid confusion, we created a new file consisting of the cleaned data.

Exploratory Analysis

We analyzed the summary statistics for temperature, dissolved oxygen, and pH. Our criterion was different locations in Clear Lake. Since the researchers were particularly concerned about Oaks Arm, we were not surprised to find that Oaks Arm contained the most observations.

Figure 1: Number Of Observations At Different Locations
Original Data From 2041 Observations

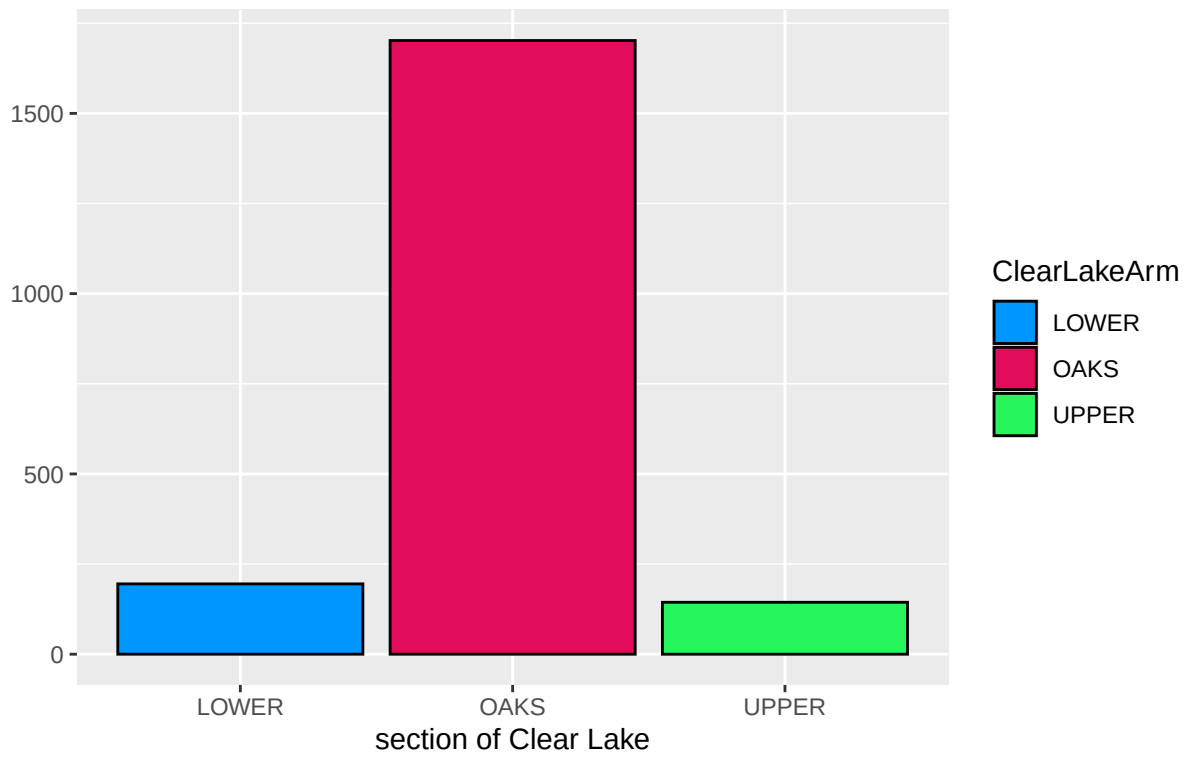
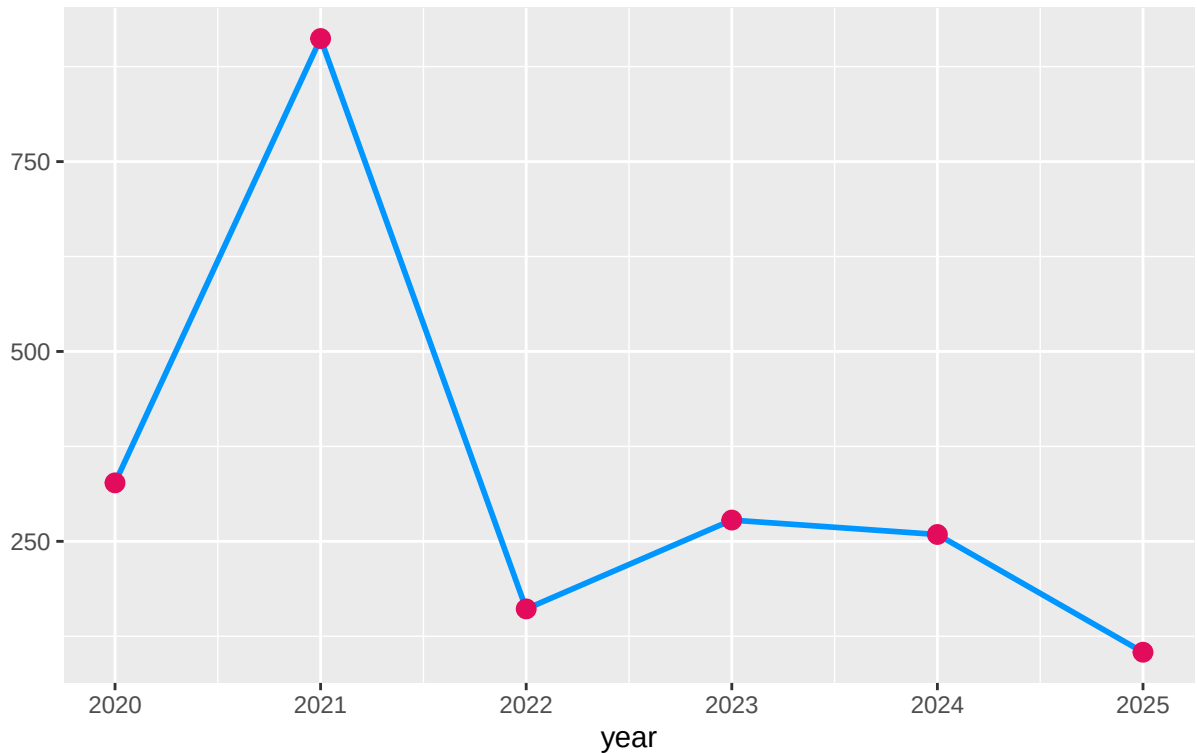


Figure 2: Number Of Observations In Different Years
Original Data From 2041 Observations



[1] "Figure 3: Summary Statistics Of Temperature(Celsius degrees) At Different Locations"

```
## # A tibble: 3 x 7
##   ClearLakeArm min   max mean median   sd   skew
##   <fct>         <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 LOWER         19.7  27.0  22.9  23.2  1.83 -0.109
## 2 OAKS           8.65  28.3  19.7  20.7  6.07 -0.464
## 3 UPPER         20.4  27.6  22.9  22.8  1.59  0.567
```

[1] "Figure 4: Summary Statistics Of Dissolved Oxygen(mg/L) At Different Locations"

```
## # A tibble: 3 x 7
##   ClearLakeArm min   max mean median   sd   skew
##   <fct>         <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 LOWER          0.15  14.6  6.55  6.41  3.20  0.0462
## 2 OAKS          -0.07  16.0  7.68  7.70  3.36 -0.162
## 3 UPPER          0.09  11.9  6.14  5.37  3.11  0.181
```

[1] "Figure 5: Summary Statistics Of pH At Different Locations"

```
## # A tibble: 3 x 7
##   ClearLakeArm min   max mean median   sd   skew
##   <fct>         <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 LOWER          7.22  9.08  8.25  8.31  0.463 -0.332
## 2 OAKS           6.12  9.3   8.17  8.29  0.623 -0.808
## 3 UPPER          7.34  9.01  8.29  8.34  0.392 -0.502
```

Advanced Analysis

First Null hypothesis: The means of dissolved oxygen among three locations are not different.

First Alternative hypothesis: At least one location carries a different mean of dissolved oxygen.

We hoped to know whether the mean of dissolved oxygen is significantly different among Upper Arm, Lower Arm, and Oaks Arm. We launched an ANOVA test. Since the p-value was very small, we could reject the null hypothesis. Furthermore, we added a Tukey's HSD test. Since the p-value involving Oaks Arm was small, we were confident to say that the level of dissolved oxygen in Oaks Arm was distinct. The proximity to mercury mine might explain this phenomenon.

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## ClearLakeArm    2     489   244.30   22.06 3.37e-10 ***
## Residuals     1938   21463    11.07
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 100 observations deleted due to missingness

##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = EX02_Field_DO_.mg.L. ~ ClearLakeArm, data = outlook_new)
##
## $ClearLakeArm
##              diff              lwr              upr              p adj
## OAKS-LOWER    1.1282615  0.5362484  1.7202746  0.0000246
## UPPER-LOWER  -0.4101506 -1.2677949  0.4474936  0.5007678
## UPPER-OAKS   -1.5384121 -2.2174836 -0.8593407  0.0000004
```

Second Null hypothesis: The level of dissolved oxygen is not relevant to pH.

Second Alternative hypothesis: There is an association between the level of dissolved oxygen and pH.

Adopting all observations, we constructed a linear model. One mg/L increase in dissolved oxygen might lead to 0.093051 increase of pH. About 30.18 percent of all variations in pH could be explained by variations in dissolved oxygen. Furthermore, since the p-value was very small, we could reject the null hypothesis.

```
##
## Call:
## lm(formula = EX02_Field_pH_.std_units. ~ EX02_Field_DO_.mg.L.,
##     data = outlook_new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.76442 -0.29887 -0.01552  0.37440  1.39191
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.499683   0.026421  283.86  <2e-16 ***
## EX02_Field_DO_.mg.L. 0.093051   0.003227   28.84  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4749 on 1921 degrees of freedom
```

```
## (118 observations deleted due to missingness)
## Multiple R-squared:  0.3021, Adjusted R-squared:  0.3018
## F-statistic: 831.6 on 1 and 1921 DF,  p-value: < 2.2e-16
```

Summary and Conclusion

Our analysis indicated that Oaks Arm was somehow prominent in the level of pollution. Moreover, different indicators of pollution might have correlation with each other.

A key limitation was that the abundance of missing values undermined the effectiveness of our analysis. In addition, the time distribution of the original data was uneven. Therefore, we encountered some obstacles in more advanced time series analysis. If we had got more time, we would have completed our stratified analysis on year and tested multiple regression models.

References

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